

MONTREAL ST-HUBERT, QC, Canada

WMO: 713710

Lat: 45.520N

Lon: 73.420W

Elev: 90

StdP: 14.65

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.8	(c) -5.0	(d) -20.6	(e) 1.8	(f) -8.5	(g) -15.8	(h) 2.3	(i) -4.1	(j) 31.8	(k) 27.3	(l) 28.5	(m) 23.5	(n) 8.3	(o) 270	(p) 0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.6	(c) 86.4	(d) 72.0	(e) 83.7	(f) 70.2	(g) 81.2	(h) 68.8	(i) 74.6	(j) 82.7	(k) 72.7	(l) 80.2	(m) 71.1	(n) 78.1	(o) 13.1	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 71.9	(b) 118.4	(c) 79.1	(d) 70.2	(e) 111.4	(f) 77.3	(g) 68.5	(h) 105.1	(i) 75.7	(j) 38.2	(k) 82.7	(l) 36.5	(m) 80.3	(n) 35.0	(o) 78.4	(p) 82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	22.4	20.0	-15.6	90.9	6.1	2.9	-20.0	93.0	-23.6	94.7	-27.0	96.3	-31.4	98.4
			-16.2	78.0	5.9	2.1	-20.4	79.6	-23.8	80.8	-27.1	82.0	-31.4	83.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	15.8	18.7	28.7	42.8	56.5	65.6	70.4	68.4	60.8	48.2	35.9	23.6		
	DBStd	20.97	12.66	10.82	10.28	8.29	7.57	6.63	5.11	5.37	7.32	7.94	9.00	11.03		
	HDD50	4249	1060	876	663	243	23	0	0	0	4	133	428	819		
	HDD65	7850	1525	1296	1126	666	280	72	10	27	168	523	873	1284		
	CDD50	2333	0	0	2	28	225	468	631	571	327	76	5	0		
	CDD65	460	0	0	0	1	17	90	177	134	40	1	0	0		
	CDH74	3437	0	0	1	16	193	712	1306	914	287	8	0	0		
	CDH80	794	0	0	0	2	39	181	322	192	58	0	0	0		
(14) Wind	WSAvg	9.5	10.8	10.5	10.4	10.6	9.7	8.7	8.4	7.9	8.2	9.2	9.8	10.0		
(15) Precipitation	PrecAvg	40.2	3.0	2.3	2.6	3.3	3.5	3.9	4.0	3.8	3.5	3.9	3.0	3.5		
	PrecMax	49.8	5.6	4.1	4.7	6.0	7.2	7.7	6.2	7.0	7.8	7.2	5.5	6.5		
	PrecMin	32.4	1.1	0.9	0.7	0.7	1.7	1.8	2.1	1.5	1.0	1.2	0.5	1.1		
	PrecStd	4.3	1.3	0.8	1.0	1.5	1.3	1.4	1.0	1.3	1.5	1.6	1.2	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.8	44.9	60.1	75.6	84.6	88.9	90.2	88.2	86.2	74.7	62.6	51.7		
		MCWB	46.2	41.3	50.7	61.6	66.9	72.8	74.5	73.4	72.5	64.5	57.3	48.2		
	2%	DB	41.2	40.7	51.2	67.6	79.3	84.9	86.4	84.6	80.8	69.4	57.4	45.5		
		MCWB	39.2	37.5	44.2	54.3	63.2	70.8	72.4	71.1	69.2	61.2	53.3	43.2		
	5%	DB	37.4	37.5	46.1	61.9	74.9	81.5	83.5	81.8	77.0	64.8	52.9	41.3		
		MCWB	35.5	34.7	40.8	50.8	60.6	68.2	70.7	69.4	67.1	57.6	48.7	38.8		
	10%	DB	34.0	34.9	42.1	56.8	70.6	78.1	80.9	79.2	73.4	60.7	49.0	37.7		
		MCWB	32.2	32.9	37.8	47.4	59.1	66.2	69.3	68.3	65.3	54.8	45.1	35.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.6	42.5	53.3	62.8	70.0	76.3	77.5	76.0	74.1	66.5	58.6	49.3		
		MCDB	47.4	44.9	58.7	72.1	81.6	85.8	87.5	84.7	81.8	72.3	61.1	51.4		
	2%	WB	39.6	37.8	45.2	56.8	66.5	73.0	74.9	73.5	71.2	62.0	53.7	43.4		
		MCDB	40.9	40.2	49.9	64.6	74.9	81.7	83.0	81.0	77.9	67.6	57.5	45.3		
	5%	WB	35.6	35.1	41.3	52.0	63.7	70.5	72.9	71.7	68.7	58.5	49.4	39.2		
		MCDB	37.5	37.0	45.2	59.7	71.4	77.9	80.1	78.6	75.4	63.9	52.2	41.1		
	10%	WB	32.3	33.0	38.2	48.6	60.6	68.4	71.2	70.0	66.2	55.6	45.5	35.9		
		MCDB	34.0	34.9	41.6	55.7	67.9	75.6	78.2	76.5	72.4	59.8	48.6	37.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	14.7	14.9	17.8	19.4	18.2	17.6	18.2	18.7	15.5	13.1	12.5		
		MCDBR	16.2	15.7	19.7	27.4	25.6	22.3	20.2	20.4	21.8	21.5	18.6	14.5		
		MCWBR	15.7	14.4	14.4	17.1	13.7	11.7	10.2	10.2	12.1	14.1	15.4	14.0		
	5% WB	MCDBR	16.4	14.7	18.3	23.4	21.2	18.7	17.5	17.7	18.9	19.1	17.7	14.5		
		MCWBR	15.9	13.9	14.5	16.3	13.5	11.5	10.4	10.3	12.0	13.8	15.8	14.2		
(40) Clear-Sky Solar Irradiance	taub		0.260	0.290	0.332	0.365	0.392	0.440	0.432	0.417	0.376	0.336	0.308	0.296		
	taud		2.350	2.252	2.289	2.349	2.330	2.203	2.230	2.268	2.370	2.486	2.481	2.297		
	Ebn at Noon		272	282	284	284	278	264	264	264	266	264	252	240		
	Edh at Noon		23	32	36	37	39	44	43	40	33	25	21	23		
(44) All-Sky Solar Radiation	RadAvg		459	760	1152	1382	1623	1786	1794	1586	1257	724	438	317		
	RadStd		51	79	94	162	153	135	133	74	103	64	57	27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	N/A

Nomenclature: See separate page